

Unit - II

Routing Protocols

(Marks 16)

- Router architecture, routing table, queueing and switching
- Routing protocols- Intra domain routing- Distance vector routing-Creating distance vector routing tables, Initialization, Sharing, Updating- Routing Information Protocol (RIPv2), Link State Routing Open Shortest Path First (OSPF)-Types of links, Graphical representation, Inter domain Routing-Path Vector Routing- Border Gateway Protocol (BGPv4)
- Internet Control Message Protocol (ICMP)-Types of messages, Message format, Error reporting messages

Router Architecture

A router is a network device that connects multiple networks and directs data packets between them. Its architecture consists of:

1. **Input Ports**
 - Receives incoming data packets from different networks.
 - Performs error checking, packet classification, and buffering.
2. **Routing Processor**
 - Maintains the routing table and makes forwarding decisions.
 - Handles control plane functions like routing protocols (e.g. OSPF, BGP).
3. **Switching Fabric**
 - Connects input ports to output ports.
 - Transfers packets quickly from one port to another.
 - Can be implemented using buses, shared memory, or crossbar switches.
4. **Output Ports**
 - Queues packets before sending them to the next hop.
 - Handles forwarding, error detection, and flow control.

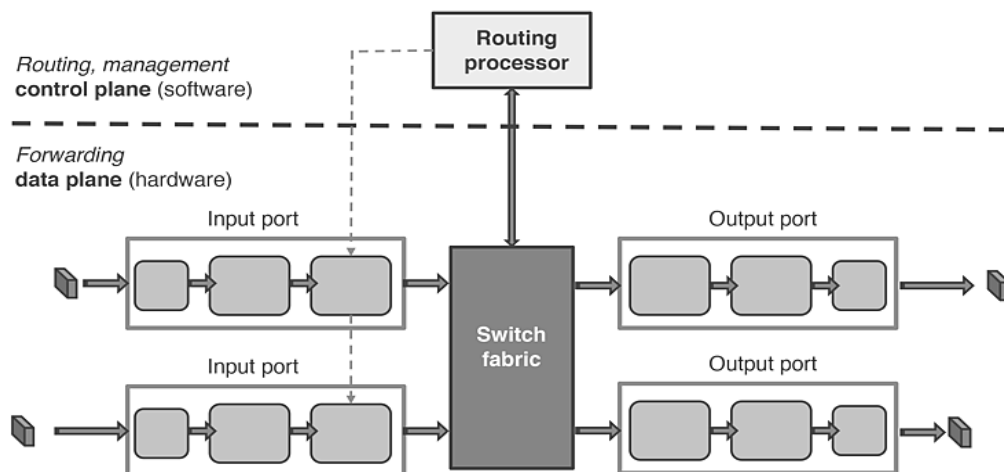


Fig. Router Architecture

Routing Table

The routing table stores information about how to reach different network destinations.

It typically includes:

- **Destination Network** – The target network or IP address.
- **Next Hop** – The address of the next router to which the packet should be sent.
- **Metric** – Cost or distance associated with reaching the destination (lower is better).
- **Interface** – The output port through which the packet should be forwarded.

Queueing in Routers

Queueing refers to the process of temporarily storing packets in memory buffers before they are transmitted over an output link. It is an essential function in routers because packets do not always arrive at the same rate as they can be forwarded. If too many packets arrive at once, they need to wait their turn before being sent, which is managed through queues.

Switching in Routers

Switching is the process by which a router transfers incoming packets from the input port to the appropriate output port so that they can be sent toward their destination. It plays a crucial role in ensuring that packets move efficiently and quickly through the router.

Switching is performed using the switching fabric, which connects the input ports and output ports and allows packets to be forwarded based on decisions made by the routing processor.

Autonomous System (AS)

- An autonomous system (AS) is a group of networks and routers under the authority of a single administration.
- Routing **inside an autonomous system** is referred to as **intra-domain routing**.
- Routing **between autonomous systems** is referred to as **inter-domain routing**.
- Each autonomous system can choose one or more intra-domain routing protocols to handle routing inside the autonomous system. However, only one inter-domain routing protocol handles routing between autonomous systems.

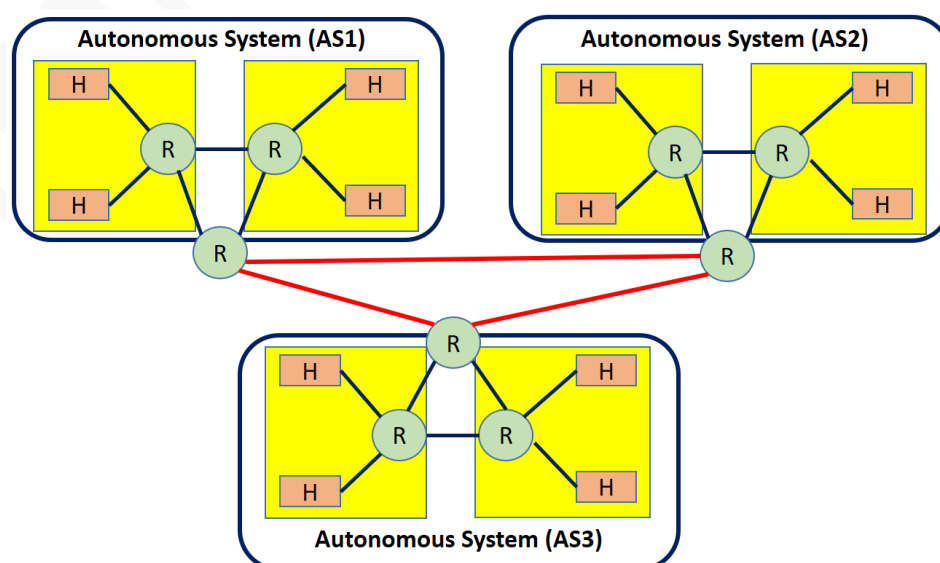


Fig. 2.1 Autonomous System

Intra-domain routing

Routing **inside an autonomous system** is referred to as **intra-domain routing**. There are different protocols for routing inside an autonomous system. These protocols are based on Intra-domain routing algorithms.

Intra-domain routing algorithm

- Distance Vector Routing algorithm
Routing Information Protocol (RIP) is based on this algorithm.
- Link State Routing algorithm
Open Shortest Path First (OSPF) is based on this algorithm.

Distance Vector Routing algorithm (Intra-domain Routing Algorithm)

- Autonomous System considered as a graph, in which each router is node and these nodes connected via links – every pair of node have a cost, represented above a link. (refer fig. 2.2)
- In distance vector routing, the ***least-cost route between any two nodes is the route*** with minimum distance.
- In this algorithm, as the name implies, each node maintains a vector (table) of minimum distances to every node.
- The table at each node also guides the packets to the desired node by showing the next stop in the route (next-hop routing).
- Steps to build Routing Table for each node.
 - Initialization
 - Sharing
 - Updating

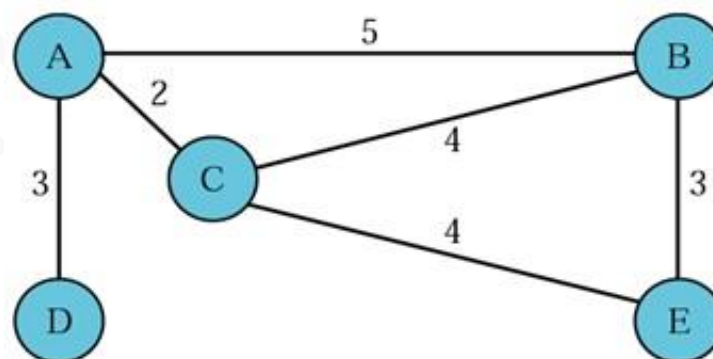


Fig. 2.2 Autonomous System (Represented as a graph)

Consider above network in fig 2.2 of nodes (A,B,C,D,E) with cost. We will apply initialization step to this network.

Initialization

At the beginning, however, this is not the case. Each node can know only the distance between itself and its immediate neighbours, those directly connected to it. So for the moment, we assume that each node can send a message to the immediate neighbours and find the distance between itself and these neighbours. Figure 3.2 shows initial vector table for each node. The distance for any entry that is not a neighbour is marked as infinite (unreachable).

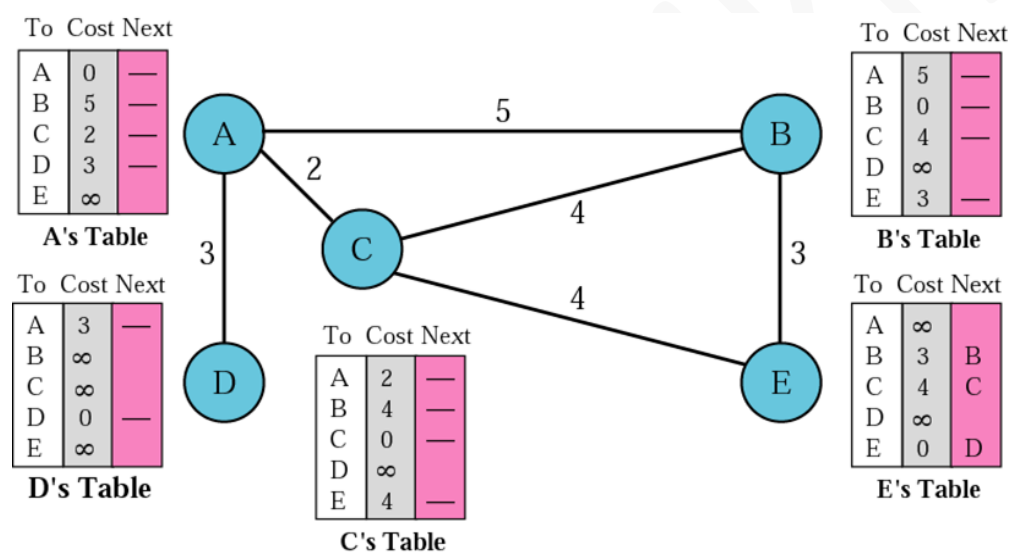


Fig. 2.3 Initial Vector tables for nodes

Sharing

In distance vector routing, each node shares its routing table with its immediate neighbours periodically and when there is a change. The whole idea of distance vector routing is the sharing of information between neighbours. Although node A does not know about node E, node C does. Therefore, if node C shares its routing table with A, node A can also know how to reach node E.

Updating in distance vector routing

When a node receives a two-column table from a neighbour, it needs to update its routing table. Updating takes three steps:

1. The receiving node needs to add the cost between itself and the sending node to each value in the second column. The logic is clear.

If node C claims that its distance to a destination is x , and the distance between A and C is y , then the distance between A and that destination, via C, is $x + y$.

2. The receiving node needs to add the name of the sending node to each row as the third column if the receiving node uses information from any row. The sending node is the next node in the route.
3. The receiving node needs to compare each row of its old table with the corresponding row of the modified version of the received table.
 - a) If the next-node entry is different, the receiving node chooses the row with the smaller cost. If there is a tie, the old one is kept.
 - b) If the next-node entry is the same, the receiving node chooses the new row. For example, suppose node C has previously advertised a route to node X with distance 3. Suppose that now there is no path between C and X; node C now advertises this route with a distance of infinity. Node A must not ignore this value even though its old entry is smaller. The old route does not exist any more. The new route has a distance of infinity.

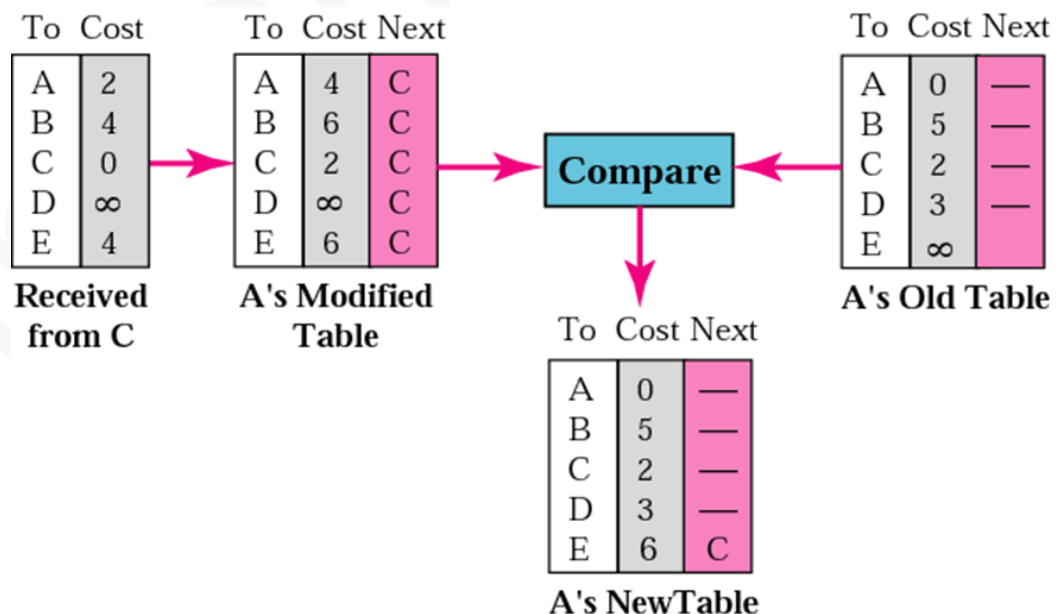


Fig. 2.4 Example for Updating Vector Table for Node A

Routing Information Protocol (RIP v2)

- RIP protocol is based on Distance vector routing algorithm.
- RIP is an Intra-domain routing protocol.
- Maximum Hop count limit is **15**, if reached to 16 it will be considered as **Unreachable**.
- RIP suitable for small networks due to limited maximum hop count.
- RIP uses UDP service at transport layer.
- Each router sends the routing update out of all active interfaces every **30 seconds**.
- Each router also receives routing updates from its neighbouring routers.
- To send

RIP message format (RIP v2)

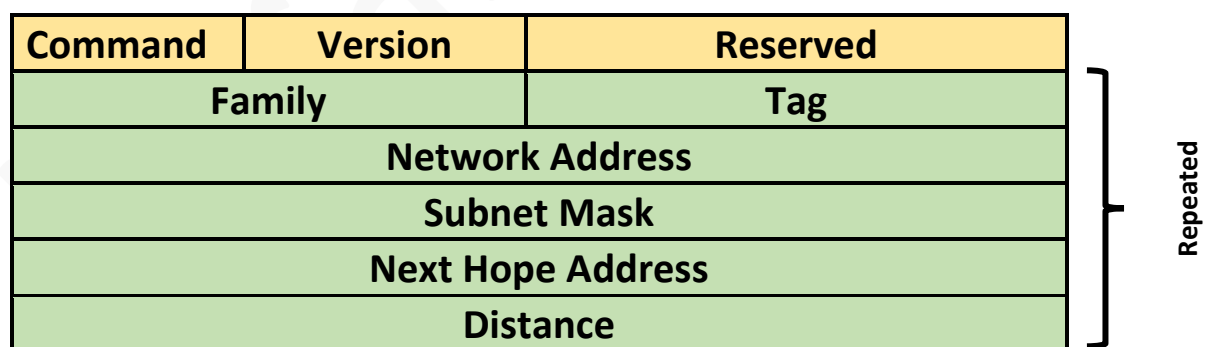


Fig. 2.6 RIP Packet format version2

Command (8bit) - type of message: request (1) or response (2).

Version (8bit) - current Version 2

Family (16bit) - Family of protocol, for TCP/IP value is 2

Tag (16bit) - Information about autonomous system

Network Address (32bit) - Destination address

Subnet Mask (32bit) – To identify prefix bits

Next Hope Address (32bit) – Defines via which interface packet can reach destination

Distance (32bit) - Number of hops to the destination

LINK STATE ROUTING - Building Routing Tables

In link state routing, four sets of actions are required to ensure that each node has the routing table showing the least-cost node to every other node.

1. Creation of the states of the links by each node, called the link state packet (LSP).
2. Dissemination of LSPs to every other router, called flooding, in an efficient and reliable way.
3. Formation of a shortest path tree for each node.
4. Calculation of a routing table based on the shortest path tree.

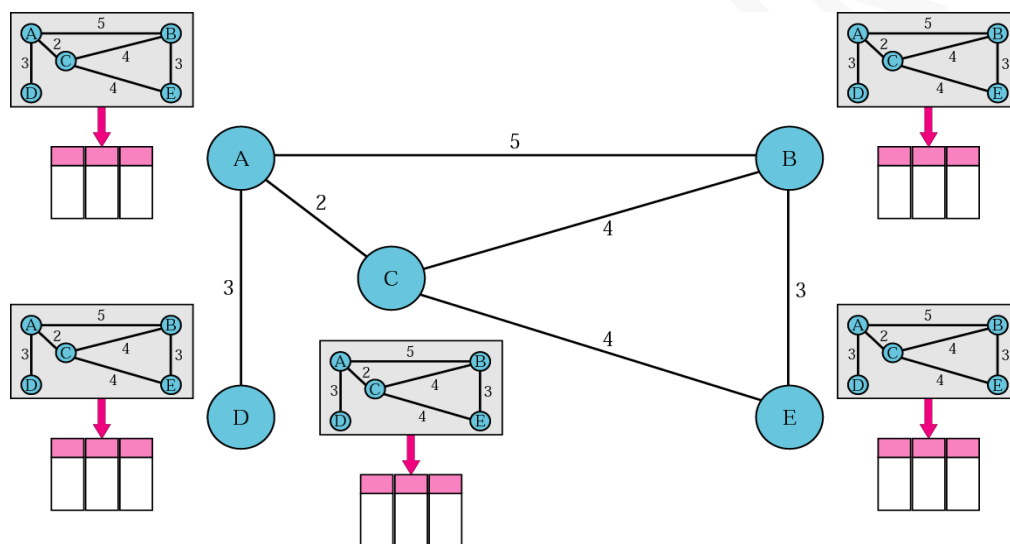


Fig. 2.7 Concept of Link State Routing (Every node have domain topology)

Creation of LINK STATE PACKET (LSP)

- A link state packet can carry a large amount of information. For the moment, however, we assume that it carries a minimum amount of data:

the node identity, the list of links, a sequence number, and age.

- The first two, node identity and the list of links, are needed to make the topology.
- The third, sequence number, facilitates flooding and distinguishes new LSPs from old ones.
- The fourth, age, prevents old LSPs from remaining in the domain for a long time.

Flooding of LSPs (Help to know complete Topology of N/W)

Flooding of LSPs After a node has prepared an LSP, it must be disseminated to all other nodes, not only to its neighbours. The process is called flooding and based on the following:

1. The creating node sends a copy of the LSP out of each interface.
2. A node that receives an LSP compares it with the copy it may already have. If the newly arrived LSP is older than the one it has (found by checking the sequence number), it discards the LSP. If it is newer, the node does the following:
 - a. It discards the old LSP and keeps the new one.
 - b. It sends a copy of it out of each interface except the one from which the packet arrived. This guarantees that flooding stops somewhere in the domain (where a node has only one interface).

Formation of Shortest path tree

A shortest path tree is a tree in which the path between the root and every other node is the shortest. What we need for each node is a shortest path tree with that node as the root.

The Dijkstra algorithm creates a shortest path tree from a graph. The algorithm divides the nodes into two sets: tentative and permanent.

It finds the neighbours of a current node, makes them tentative, examines them, and if they pass the criteria, makes them permanent. We can informally define the algorithm by using the flowchart in figure

Let us apply the algorithm to node A of our sample graph in Figure 2.8

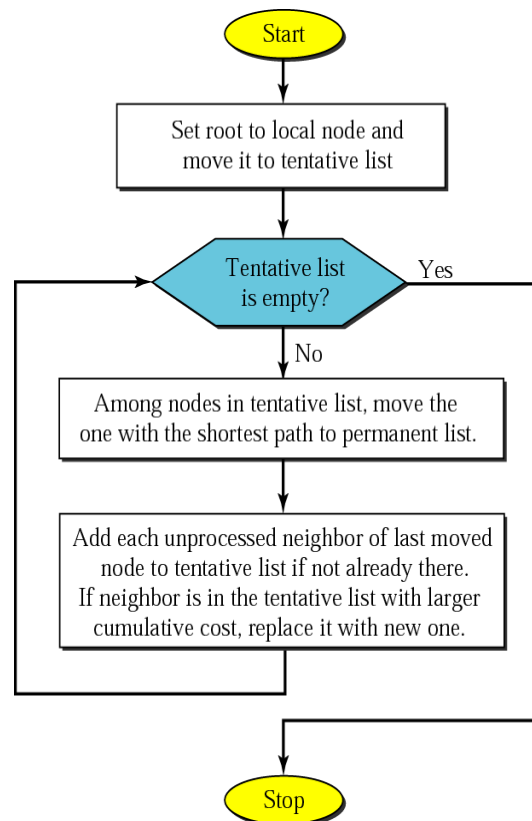


Fig. 2.8 Dijkstra Algorithm for Shortest path tree

Example to form Shortest path tree for Node A (refer a figure 2.9)

1. We make node A the root of the tree and move it to the tentative list. Our two lists are

Permanent list: empty

Tentative list: A(0)

2. Node A has the shortest cumulative cost from all nodes in the tentative list. We move A to the permanent list and add all neighbours of A to the tentative list. Our new lists are

Permanent list: A(0)

Tentative list: B(5), C(2), D(3)

3. Node C has the shortest cumulative cost from all nodes in the tentative list. We move C to the permanent list. Node C has three neighbours(A,B,E), but node A (self node) is already processed, which makes the unprocessed neighbours just B and E. However, B is already in the tentative list with a cumulative cost of 5. Node A could also reach node B through C with a cumulative cost of 6. Since 5 is less than 6, we keep node B with a cumulative cost of 5 in the tentative list and do not replace it. Our new lists are

Permanent list: A(0), C(2)

Tentative list: B(5), D(3), E(6)

4. Node D has the shortest cumulative cost of all the nodes in the tentative list. We move D to the permanent list. Node D has no unprocessed neighbour to be added to the tentative list. Our new lists are

Permanent list: A(0), C(2), D(3)

Tentative list: B(5), E(6)

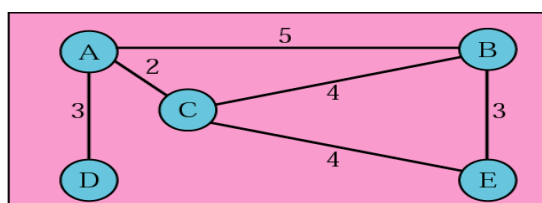
5. Node B has the shortest cumulative cost of all the nodes in the tentative list. We move B to the permanent list. We need to add all unprocessed neighbours of B to

the tentative list (this is just node E). However, E(6) is already in the list with a smaller cumulative cost. The cumulative cost to node E, as the neighbour of B, is 8. We keep node E(6) in the tentative list. Our new lists are

Permanent list: A(0), B(5), C(2), D(3) Tentative list: E(6)

6. Node E has the shortest cumulative cost from all nodes in the tentative list. We move E to the permanent list. Node E has no neighbour. Now the tentative list is empty. We stop; our shortest path tree is ready. The final lists are

Permanent list: A(0), B(5), C(2), D(3), E(6) Tentative list: empty



Topology

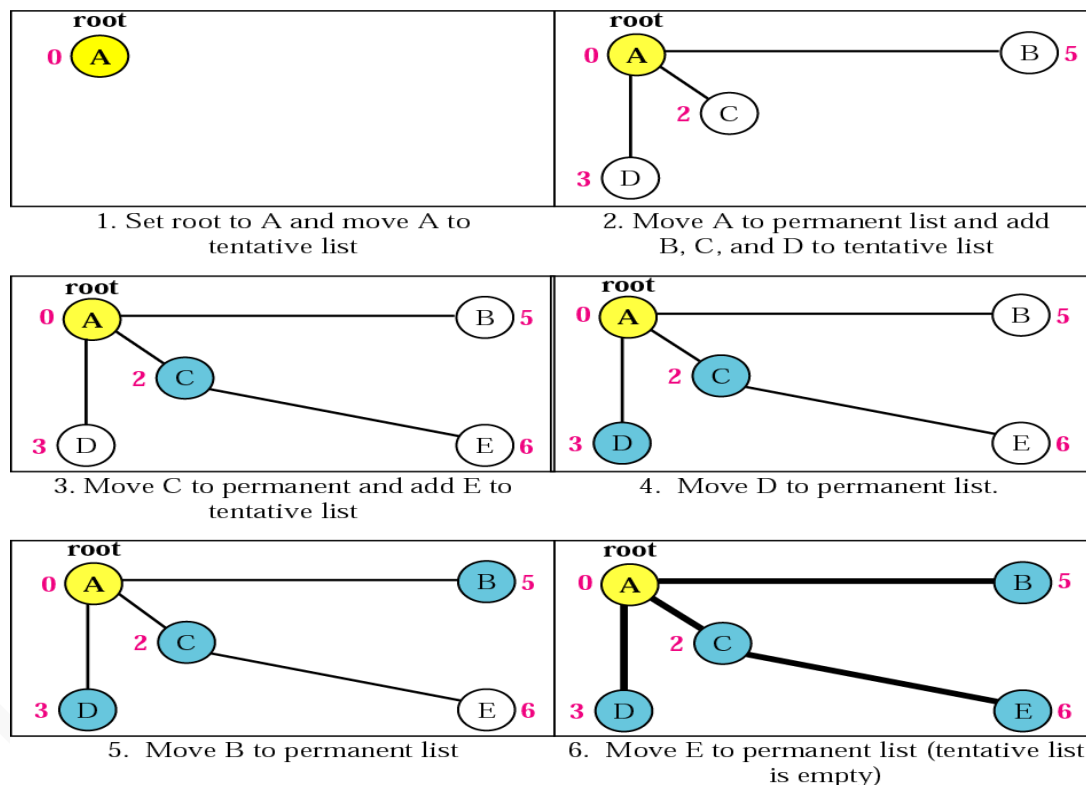


Fig. 2.9 Example of formation of shortest path tree

Calculation of Routing Table

Calculation of Routing Table from Shortest Path Tree Each node uses the shortest path tree protocol to construct its routing table. The routing table shows the cost of reaching each node from the root. Table 2.1 shows the routing table for node A.

<i>Node</i>	<i>Cost</i>	<i>Next Router</i>
A	0	—
B	5	—
C	2	—
D	3	—
E	6	C

Table 2.1

Open Shortest Path First (OSPF)

- It is an interior gateway protocol that has been designed within a single autonomous system.
- It is based on a link-state routing algorithm in which each router contains the information of every domain, and based on this information, it determines the shortest path.
- Every router contains the same information about the network. The way the router learns this information by sending LSA (Link State Advertisements).
- Once the LSAs have been flooded, the OSPF stores the information in a link-state database known as LSDB.

Areas

- Compared with RIP, which is normally used in small ASs, OSPF was designed to be able to handle routing in a small or large autonomous system. However, the formation of shortest-path trees in OSPF requires that all routers flood the whole AS with their LSPs to create the global LSDB.
- It may have created a huge volume of traffic in a large AS. To prevent this, the AS needs to be divided into small sections called areas.
- However, each router in an area needs to know the information about the link states not only in its area but also in other areas. For this reason, one of the areas in the AS is designated as the backbone area, responsible for connecting the areas together.
- By using Backbone network, a router in an area can receive all LSPs generated in other areas.

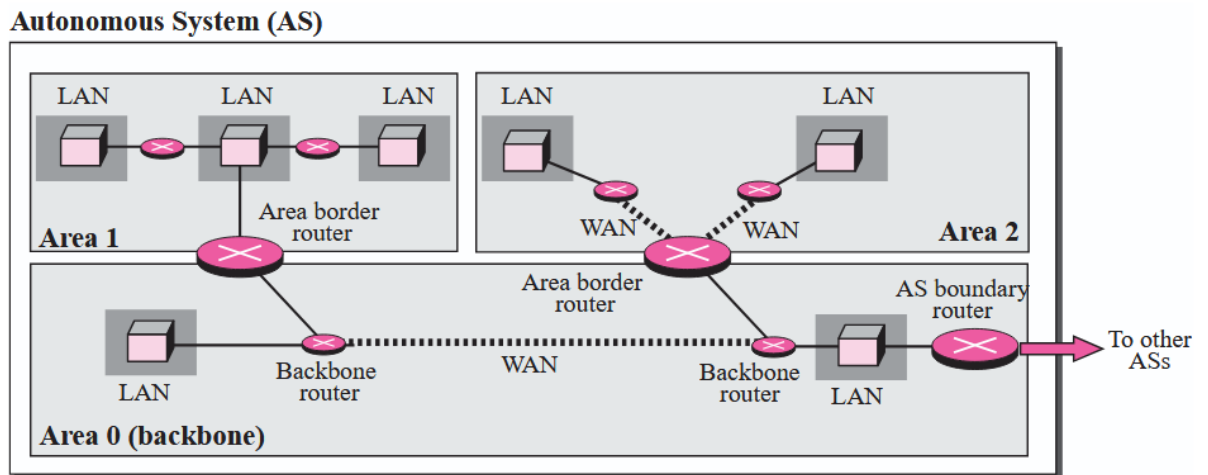


Fig. 2.10 Areas in an autonomous system

Types of Links

- Point to Point
- Transient
- Stub
- Virtual

Point to Point

A point-to-point link connects two routers without any other host or router in between.

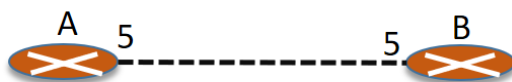


Fig. 2.11a Point to Point Network

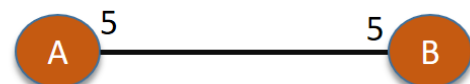


Fig.2.11b Graphical Representation

Transient link

A transient link is a *network*, where multiple routers are connected to the same physical network (like Ethernet). The data can enter through any of the routers and leave through any router.

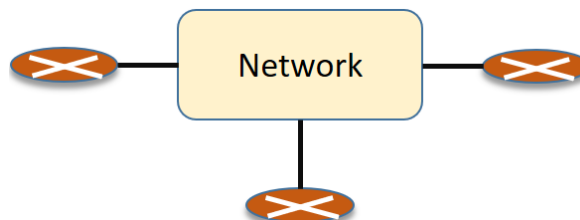


Fig. 2.12 Transient Link

Stub Link

A network segment with only one router connected. Traffic can only enter or leave through that single router. There are no OSPF neighbors on the other side.

Example: A router's loopback interface or a network connected to just one router.

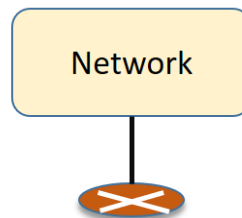


Fig. 2.13 Stub Link

Virtual Link

In OSPF, all areas must connect to Area 0 (the backbone area). Sometimes, due to network design or changes, an area gets separated from Area 0. In such a situation, the administration may create a virtual link between separated area and Area 0 (the backbone area) by using a longer path that probably goes through several routers.

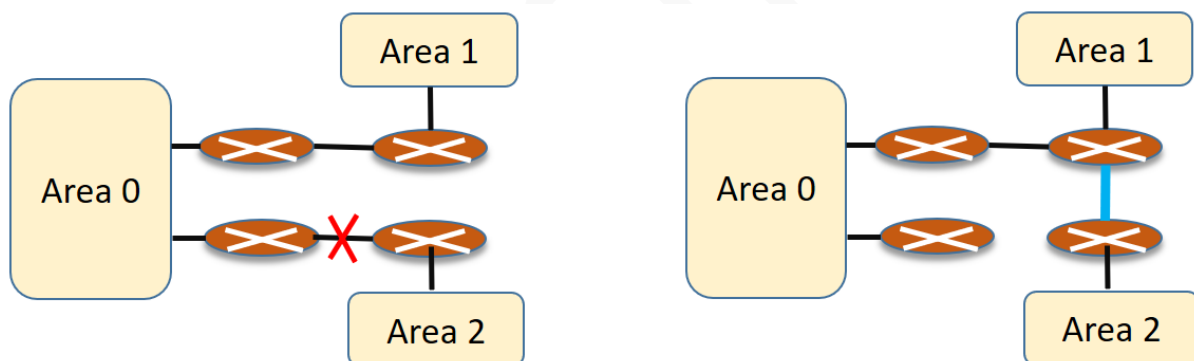


Fig. 2.14 Virtual Link

Path Vector Routing (Inter-domain Routing Algorithm)

- Path vector routing is used to route packets from one Autonomous System to another.
- The principle of path vector routing is similar to that of distance vector routing (with steps Initialization, Sharing, Update). In path vector routing, we assume that there is one node in each autonomous system that acts on behalf of the entire autonomous system. Called as **speaker node**.
- The speaker node in an AS creates an initial routing table and advertises it to speaker nodes in the neighbouring ASs. The idea is the same as for distance vector routing except that only speaker nodes in each AS can communicate with each other.

- However, what is advertised is different. A speaker node **advertises the path, not the metric (distance)** of the nodes, in its autonomous system or other autonomous systems.
- We can summarize the fields that advertise by DVR and PVR as
 - DVR - *Destination, Cost, Next Hope IP*
 - PVR - *Destination, Path(based on policies), Next Hope IP*

Initialization

At the beginning, each speaker node can know only the reachability of nodes inside its autonomous system. Figure 3.7 shows the initial tables for each speaker node in a system made off our ASs.

Sharing

A path vector routing table for each router can be created if ASs share their reachability list with each other. In Figure 2.15, router R1 in AS1 can send its reachability list to router R2. Router R2, after combining its reachability list, can send the result to both R1 and R3. Router R3 can send its reachability list to R2, which in turn improves its routing table, and so on.

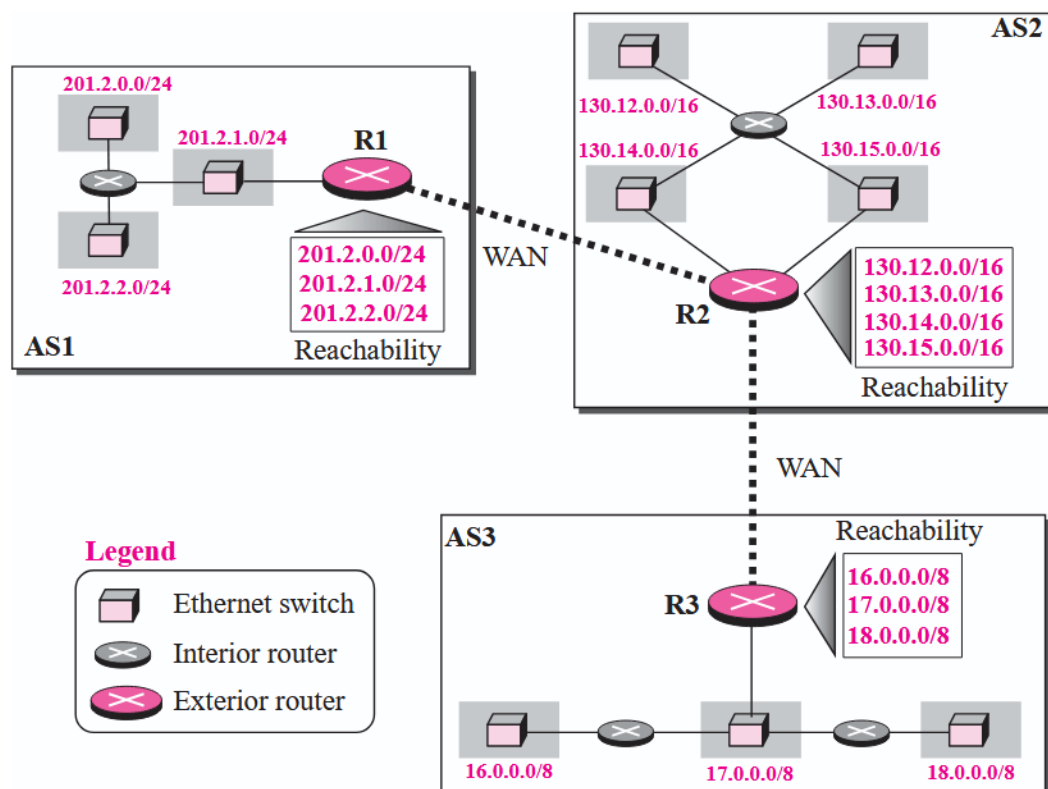


Fig. 2.15 PVR - Initialization

Updating

Figure 2.16 shows the routing table for each router after all three routers have updated their routing table.

R1		R2		R3	
Network	Path	Network	Path	Network	Path
201.2.0.0/24	AS1 (This AS)	201.2.0.0/24	AS2, AS1	201.2.0.0/24	AS3, AS2, AS1
201.2.1.0/24	AS1 (This AS)	201.2.1.0/24	AS2, AS1	201.2.1.0/24	AS3, AS2, AS1
201.2.2.0/24	AS1 (This AS)	201.2.2.0/24	AS2, AS1	201.2.2.0/24	AS3, AS2, AS1
130.12.0.0/16	AS1, AS2	130.12.0.0/16	AS2 (This AS)	130.12.0.0/16	AS3, AS2
130.13.0.0/16	AS1, AS2	130.13.0.0/16	AS2 (This AS)	130.13.0.0/16	AS3, AS2
130.14.0.0/16	AS1, AS2	130.14.0.0/16	AS2 (This AS)	130.14.0.0/16	AS3, AS2
130.15.0.0/16	AS1, AS2	130.15.0.0/16	AS2 (This AS)	130.15.0.0/16	AS3, AS2
16.0.0.0/8	AS1, AS2, AS3	16.0.0.0/8	AS2, AS3	16.0.0.0/8	AS3 (This AS)
17.0.0.0/8	AS1, AS2, AS3	17.0.0.0/8	AS2, AS3	17.0.0.0/8	AS3 (This AS)
18.0.0.0/8	AS1, AS2, AS3	18.0.0.0/8	AS2, AS3	18.0.0.0/8	AS3 (This AS)

Path-Vector Routing Table Path-Vector Routing Table Path-Vector Routing Table

Fig. 2.15 PVR – Updated Routing tables with Path

Border Gateway Protocol (BGP)

- It is an Exterior gateway protocol that has been designed for routing between multiple autonomous system (Inter-domain Routing).
- It is based on a path vector routing algorithm.
- It runs over TCP port number 179, support reliability.
- Separate TCP Connection is established between BGP Speaker routers to exchange Information regarding reachability to different AS.

BGP works for routing between different autonomous systems, Autonomous systems may be of different type.

Types of Autonomous System

Stub AS

It has only one connection to another AS. Data traffic cannot pass through a stub AS. A Stub AS is either a source or sink.

Multihomed AS

It has more than one connection to other ASs but it is still only a source or sink for data traffic.

Transit AS

It is a Multihomed AS that allows transient traffic.

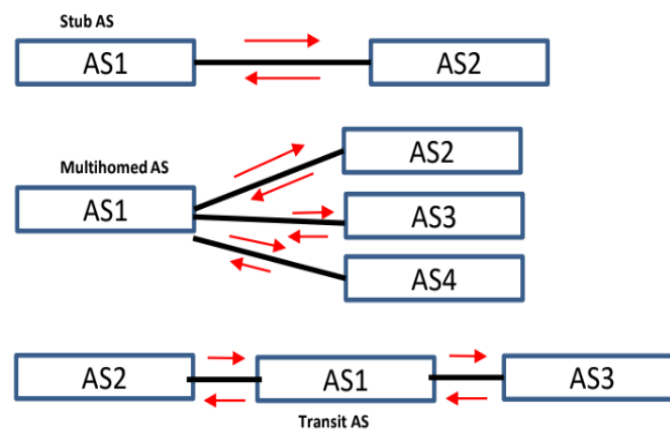


Fig. 2.16 Types of Autonomous System

Types of BGP Sessions

E - BGP:

Exchange Information between two speaker nodes

I - BGP:

Exchange information between routers of same AS.

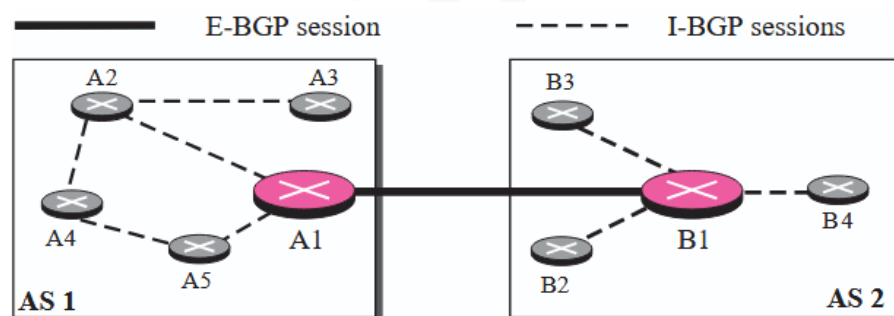


Fig. 2.17 Types of BGP Sessions

Types of BGP packets/message

Open Message

To create a neighbourhood relationship, a router running BGP opens a TCP connection with a neighbour and sends an open message.

Update Message

The update message is the heart of the BGP protocol. It is used by a router to withdraw destinations that have been advertised previously, to announce a route to a new destination, or both. Note that BGP can withdraw several destinations that were advertised before, but it can only advertise one new

destination (or multiple destinations with the same path attributes) in a single update message.

Keepalive Message

The BGP peers that are running exchange keepalive messages regularly (before their hold time expires) to tell each other that they are alive.

Notification

A notification message is sent by a router whenever an error condition is detected or a router wants to close the session.

Internet Control Message Protocol (ICMPv4)

IPv4 has no mechanism for error reporting or error correcting.

IPv4 protocol also lacks a mechanism for host and management queries.

The Internet Control Message Protocol version 4 (ICMPv4) has been designed to compensate for the above two deficiencies. It is a companion to the IP protocol.

General Format for ICMP messages

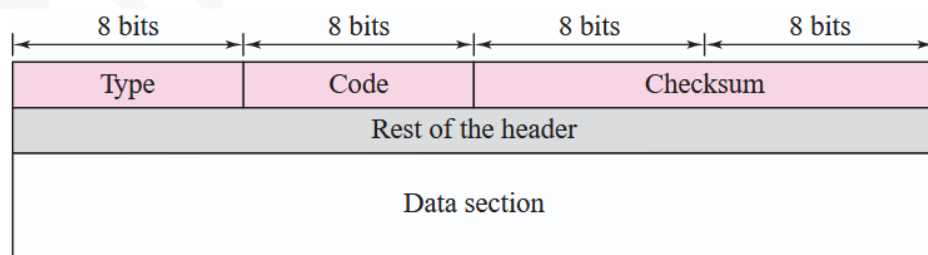


Fig. 2.18 General format for ICMP message

The first field, ICMP type, defines the type of the message. The code field specifies the reason for the particular message type. The last common field is the checksum field. The rest of the header is specific for each message type.

There are two categories of ICMP messages

Error Reporting Message

- Destination Unreachable
- Source Quench
- Time Exceeded
- Parameter problem
- Redirection

Query Message

- Echo Request and Echo Reply
- Timestamp Request and Timestamp Reply
- Router Solicitation and Advertisement
- Address Mask and Reply

Error Reporting Message

Destination Unreachable (type 3)

When a router cannot route a datagram or a host cannot deliver a datagram, the datagram is discarded and the router or the host sends a destination-unreachable message back to the source host that initiated the datagram. Note that destination unreachable messages can be created by either a router or the destination host.

Source Quench

A source-quench message informs the source that a datagram has been discarded due to congestion in a router or the destination host. The source must slow down the sending of datagram's until the congestion is relieved.

Time Exceeded

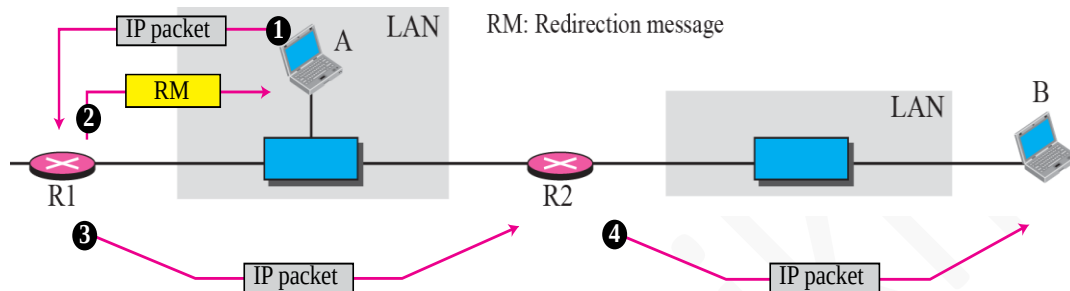
Whenever a router decrements a datagram with a time-to-live value to zero, it discards the datagram and sends a time-exceeded message to the original source.

When the final destination does not receive all of the fragments in a set time, it discards the received fragments and sends a time-exceeded message to the original source.

Parameter Problem

Any ambiguity in the header part of a datagram can create serious problems as the data-gram travels through the Internet. If a router or the destination host discovers an ambiguous or missing value in any field of the datagram, it discards the datagram and sends a parameter-problem message back to the source.

Redirect Message



- R1 will send a Redirection Message to Host A that Host A can send datagram directly to R2.
- A host usually starts with a small routing table that is gradually augmented and updated. One of the tools to accomplish this is the redirection message.

Query Messages (ICMP)

ECHO-REQUEST(type 8) and REPLY(type 0)

The echo-request and echo-reply messages can be used to determine if there is communication at the IP level. Because ICMP messages are encapsulated in IP datagram's, the receipt of an echo-reply message by the machine that sent the echo request is proof that the IP protocols in the sender and receiver are communicating with each other using the IP datagram.

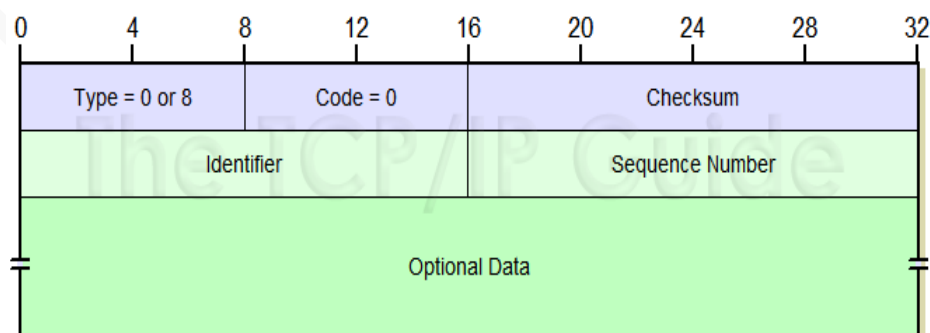


Fig 2.19 ICMP Echo Request/Reply Format

Identifier

An identifier to match Timestamp Replies to the corresponding Timestamp Requests

Sequence Number:

A sequence number to match Timestamp Replies to the corresponding Timestamp Requests

TIMESTAMP REQUEST (type 13) AND REPLY (type 14)

Two machines (hosts or routers) can use the timestamp request and timestamp reply messages to determine the round-trip time needed for an IP datagram to travel between them.

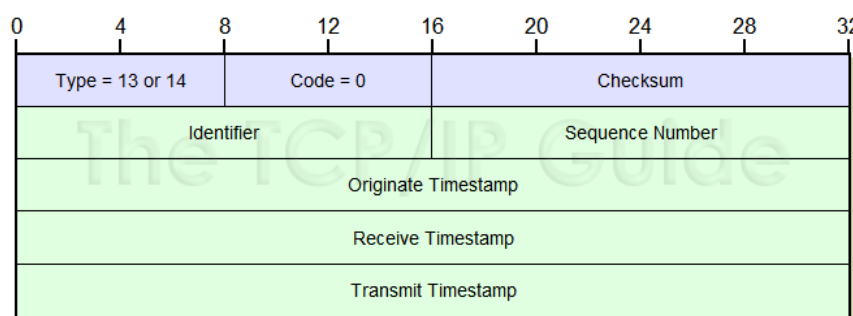


Fig 2.20 ICMP Echo Request/Reply Format

Originate Timestamp

A time value filled in by the originating device just before sending the Timestamp message.

Receive Timestamp

A time value filled in by the responding device just as it receives the Timestamp message.

Transmit Timestamp

A time value filled in by the responding device just before sending back the Timestamp Reply message.

ADDRESS MASK REQUEST (type 17) & ADDRESS MASK REPLY (type 18):

A host may be aware of its own IP address but it is not necessary that it is aware of its own subnet mask. To know your subnet mask sends the address mask request to the host router and the router sends this host's subnet mask address as the mask

reply message. If the host is aware of the router's address, then he sends the request directly to the router, otherwise this request is broadcast.

ROUTER SOLICITATION (type 10) AND ADVERTISEMENT (type 9)

As we discussed in the redirection message section, a host that wants to send data to a host on another network needs to know the address of routers connected to its own network. In addition, the host must know if the routers are alive and functioning. The router-solicitation and router-advertisement messages can help in this situation.

RIP v/s OSPF

Feature	RIP	OSPF
Algorithm	Distance Vector	Link State
Metric	Hop count	Cost (based mainly on bandwidth)
Maximum hops	15 hops	No 15-hop limitation
Convergence	Slower	Faster
Network size	Suitable for small networks	Suitable for medium/large networks
Update frequency	Typically every 30 seconds	Sends LSAs when topology changes, with periodic refreshes
Administrative Distance	120	110
Hierarchical design	No	Yes, using areas